Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		2517063

Enjoy a Cozy and Green Bath

Summary

A traditional bathtub cannot be reheated by itself, so users have to add hot water from time to time. Our goal is to establish a model of the temperature of bath water in space and time. Then we are expected to propose an optimal strategy for users to keep the temperature even and close to initial temperature and decrease water consumption.

To simplify modeling process, we firstly assume there is no person in the bathtub. We regard the whole bathtub as a thermodynamic system and introduce heat transfer formulas.

We establish two sub-models: adding water constantly and discontinuously. As for the former sub-model, we define the mean temperature of bath water. Introducing Newton cooling formula, we determine the heat transfer capacity. After deriving the value of parameters, we deduce formulas to derive results and simulate the change of temperature field via CFD. As for the second sub-model, we define an iteration consisting of two process: heating and standby. According to energy conservation law, we obtain the relationship of time and total heat dissipating capacity. Then we determine the mass flow and the time of adding hot water. We also use CFD to simulate the temperature field in second sub-model.

In consideration of evaporation, we correct the results of sub-models referring to some scientists' studies. We define two evaluation criteria and compare the two sub-models. Adding water constantly is found to keep the temperature of bath water even and avoid wasting too much water, so it is recommended by us.

Then we determine the influence of some factors: radiation heat transfer, the shape and volume of the tub, the shape/volume/temperature/motions of the person, the bubbles made from bubble bath additives. We focus on the influence of those factors to heat transfer and then conduct sensitivity analysis. The results indicate smaller bathtub with less surface area, lighter personal mass, less motions and more bubbles will decrease heat transfer and save water.

Based on our model analysis and conclusions, we propose the optimal strategy for the user in a bathtub and explain the reason of uneven temperature throughout the bathtub. In addition, we make improvement for applying our model in real life.

Keywords: Heat transfer; Thermodynamic system; CFD; Energy conservation

Contents

1 Introduction

1.1 Problem Background

The Olympic Games is one of the most influential sporting events globally. It not only showcases the outstanding talents of athletes but also promotes cultural exchanges and cooperation among nations. Since the revival of the modern Olympics in 1896, the Games has become an important platform for countries to display their athletic prowess. The medal table has always been an integral part of the Olympics, reflecting not only the sports levels of different countries but also being closely related to their economic, social, and cultural development. The number and rank of medals are often seen as one of the symbols of a country's comprehensive national strength. The medal table of the 2024 Paris Olympics shows the competitive landscape of countries in terms of gold medals and total medals. For example, the United States ranked first in total medals, while China and the United States were tied for first in gold medals. The host country, France, also performed well, ranking fourth. In addition, some countries such as Albania, Cabo Verde, Dominica, and Saint Lucia won medals for the first time in the Olympics, which reflected the diversity and inclusiveness of the Games.

(figure) (figure)

1.2 Restatement of the Problem

Question 1: Development of the Medal Quantity Prediction Model

⋆ Develop a model to predict the number of medals (at least including gold medals and total medals) that each country will win at the 2028 Los Angeles Olympics. In addition, assess the uncertainty and prediction accuracy of the model, and provide prediction intervals.

⋆ Predict which countries may improve in 2028 and which may perform worse than in 2024.

⋆ Consider countries that have not yet won medals, and predict how many countries may win medals for the first time in 2028, along with probability estimates.

⋆ The model should also explore the relationship between the setting of Olympic events and the medal-winning situations of various countries, and discuss the impact of the host country's event selection on the medal table. Identify which sports contribute the most to the medal counts of different countries and explain the reasons.

Question 2: "Great Coach" Effect Analysis

⋆ Identify the impact of "great coach" on the number of medals through data analysis.

⋆ Estimate the proportion of this effect on the number of medals.

⋆ Select three countries and analyze which sports they require the introduction of "great coach", and estimate the potential impact.

Question 3:Other Original Insights

⋆ Provide other original insights revealed by the model regarding the Olympic medal table.

⋆ Explain how these insights can provide decision support for country Olympic Committees.

1.3 Literature Review

1.4 Our Work

XXXXX

(figure)

2 Assumptions and Justification

To simplify the problem and make it convenient for us to simulate real-life conditions, we make the following basic assumptions, each of which is properly justified.

♦ **Assumption 1: XXXXX.**

Justification: XXXXX

♦ **Assumption 2: XXXXX.**

Justification: XXXXX

♦ **Assumption 3: XXXXX.**

Justification: XXXXX

♦ **Assumption 4: XXXXX.**

Justification: XXXXX

3 Notations

Symbols	Description	Unit
h	Convection heat transfer coefficient	W/(m$^2\cdot$ K)
k	Thermal conductivity	W/(m $\cdot$ K)
c_p	Specific heat	J/(kg $\cdot$ K)
ρ	Density	kg/m^2
δ	Thickness	m
t	Temperature	°C, K
τ	Time	s, min, h
q_m	Mass flow	kg/s
Φ	Heat transfer power	W
T	A period of time	s, min, h
V	Volume	m^3, L
M, m	Mass	kg
A	Aera	m^2
a, b, c	The size of a bathtub	m^3

where we define the main parameters while specific value of those parameters will be given later.

4 Data Pre-processing

4.1 Data Overview

Conduct a general analysis of the data as shown in Figure 1.

Figure 1: Overview of four data tables

Notably:

a) In 1936, during Nazi Germany, the Berlin Olympics was boycotted.

b) In 1980, during the Cold War, the Moscow Olympics was boycotted by the West.

c) In 1984, during the Cold War, the Soviet Union boycotted the Los Angeles Olympics in the United States.

These significant historical events caused large fluctuations in medal counts, resulting in some countries having zero medals, which can be analyzed further.

Encoding detection of the provided dataset files is shown in Table 1.

Table 1: Multicolumn table.

Value 1	**Value 2**	**Value 3**
α	β	γ
12		a
2	10.1	b
3	23.113231	c
4	25.113231	d

For convenience in subsequent processing, the above files were uniformly converted to .xlsx files in UTF-8 encoding format.

4.2 Data Cleaning

Data cleaning was performed on the four data files: "summerOly_athletes.csv", "summerOly_hosts.csv", "summerOly_medal_counts.csv", and "summerOly_programs.csv". The specific

Table 2: Data files and encoding formats

File	Encoding
data_dictionary.csv	**windows 1252**
summerOly_athletes.csv	**UTF-8**
summerOly_hosts.csv	**UTF-8 with BOM**
summerOly_medal_counts.csv	**UTF-8**
summerOly_programs.csv	**windows 1252**

steps are as follows:

4.2.1 In "summerOly_athletes.csv" file

a) There is an athlete named "671" in row 244172, which is clearly not a valid name. To facilitate subsequent unified processing, after checking other information about the athlete, the name was corrected to Liu Qinglian.

b)Duplicate value check revealed 1466 duplicate rows in the file, which have been deduplicated.

	Year	Sport	yearly_sport_total_medal_num
719	2024	3x3 Basketball, Basketball	0
735	2024	Cycling Road, Cycling Mountain Bike	0
736	2024	Cycling Road, Cycling Track	0
737	2024	Cycling Road, Triathlon	0
748	2024	Marathon Swimming, Swimming	0

Figure 2: The statistical situation of the abnormal Sport category

4.2.2 In "summerOly_programs.csv" file

a) There are encoding issues, suspected to be due to encoding problems, so it was processed using the Windows 1252 encoding.

b) There are missing values in the programs.csv file. The detailed missing situation is shown in Figure 2.

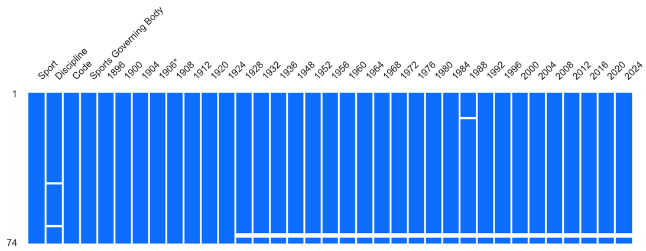


Figure 3: Schematic diagram of missing values in the programs table

i.The value for 1988 in row 14 is missing.

ii.Figure skating and ice hockey, which are ice sports, were held at the Summer Olympics before 1924. However, they were moved to the first Winter Olympics in 1924, so the values for figure skating and ice hockey in rows 71 and 72 for 1924 and later are missing.

iii.Because Modern Pentathlon and Water Motorsports do not have further classifications, the "Discipline" values are missing in rows 46 and 67. Based on other data analysis, the "Sport" values of the row was filled into the "Discipline" values.

4.2.3 In "summerOly_athletes.csv" and "summerOly_hosts.csv" files

a)Datas for the year 1906 exist in "summerOly_athletes.csv" file but not in the hosts.csv file.

△ Reason: According to research, an unofficial Olympics was held in Athens, Greece, in 1906.

b)Datas for the years 2028, 2032, 1944, 1940, 1916 exist in "summerOly_hosts.csv" file but not in "summerOly_athletes.csv" file.

△ Reason: The Olympics for the years 2028, 2032 have not yet taken place. The host countries and cities have been determined, but the participating athletes have not been determined. The three Olympics originally scheduled for the years 1944, 1940, 1916 were canceled due to the two world wars in the 20th century. Although the hosts had been determined, there was no list of participating athletes because they could not be held.

4.2.4 In "summerOly_medal_counts.csv" file

After reading with UTF-8 encoding, it was found that the 'NOC' column had trailing spaces. All strings in this column were processed to remove excess leading and trailing spaces.

5 Task 1:Development of the Medal Quantity Prediction Model

5.1 Sub-model I:A Multivariate Linear Regression-Based Model for Predicting the Total Medal Count of Countries in the Olympics

5.1.1 Model Preparation

From the systematic analysis of influencing factors in the earlier stages, we know that the competitive performance of various countries in the Olympics, specifically quantified as the proportion of the total number of medals won in Olympic competitions, is closely related to the proportion of the total number of events in which the country participates in that year's Olympics. At the same time, the past performance of the country in the Olympics, especially in the previous year and recent years, has a significant and undeniable impact on the total number of medals for the current session.

Based on these factors, let $MP^i(t)$ represent the proportion of total medals won by the i-th country in the t-th Olympics; $S-Num^i(t)$ represents the total number of events participated in by the i-th country in the t-th Olympics; $P-Num^i(t)$ is the number of medals awarded to athletes from the i-th country in the t-th Olympic Games; $Home^i(t)$is a dummy variable, when the i-th country is the host country of the t-th Olympics, $Home^i(t) = 1$, otherwise it equals 0.

5.1.2 Model Establishment

Because $MP^i(t)$ represents the medal-winning rate of each country, it takes values between 0 and 1. We perform a logarithmic transformation on $S\_Num^i(t)$ and $P\_Num^i(t)$, namely:

$S\_Num^i(t) = log(S{-}Num^i(t) + 1)$, $P\_Num^i(t) = log(P{-}Num^i(t) + 1)$. Thus, it satisfies

$$\begin{aligned} MP^i(t,\theta) &= a_1^i + a_2^i \times S\_Num^i(t) + a_3^i \times P\_Num^i(t) \\ &+ a_4^i \times Home^i(t) + a_5^i \times MP^i(t-4) + \varepsilon^i(t) \end{aligned} \tag{1}$$

Where, $\theta = [a_1, a_2, a_3, a_4, a_5]$ is the regression coefficient; t is the year of hosting the Olympics; $\varepsilon^i(t)$ encompasses all uncertain factors.

Then we fit it using the least squares method. Let $y^i(t)$ be the observed value for the i-th country in the t-year Olympics. The regression result is $\hat{MP}^i(t,\hat{\theta}) = \sum_{j=1}^{5} a_j^i \times f^i(t,j)$, and the loss function is $L(\hat{\theta}) = \sum_t [y^i(t) - \hat{M}^i(t,\hat{\theta})]^2$. The parameter estimate that minimizes $\hat{\theta}$ is taken as $L(\hat{\theta})$, thus $\hat{MP}^i(t,\hat{\theta})$ is the regression outcome.

From the analysis of the above figure, it can be seen that the total count of each Olympic Games shows a linear trend over time. It is noted that data points for the years 1916, 1940, and 1944 are missing, as during these years the Olympics were cancelled due to historical events, thus no records exist. The scatter plot above indicates a generally linear trend. Therefore, we employ a linear function to fit this trend.

$$M\_Sum = \beta_0 + \beta_1 t \tag{2}$$

Where, $\hat{\beta} = [\beta_0, \beta_1]$ is the regression coefficient; t refers to the year of the Olympics -1892.

We employ OLSE to estimate $\hat{\beta}$, and predict that the total number of medals for the years 2028 and 2032 in Table XXX.

Table 3: The prediction of the total number of medals.

Year	2028	2032
Pre_Sum	1026	1052

$Me^i(t) = Sum\_M(t) \times MP^i(t)$

5.2 Sub-model II: Exploring the Medal Distribution of Countries Based on Bayesian Models

5.2.1 Model Preparation

In the above model construction, we can predict the total medal counts for each country. Let the total number of medals be N, with X, Y, and Z representing the number of gold, silver, and bronze medals respectively, then $N = X + Y + Z$.

(Figure XXX) If we rank by scoring system, then the total score is $S = W_1 \times X + W_2 \times Y + W_3 \times Z$, with countries scoring higher ranking ahead; if ranked by total medal count, then $S = N$, with countries winning more medals ranking higher.

Based on historical data and experience regarding countries' medal counts, we know that athletes participating in the Olympics can fall into one of four categories: gold medalists, silver medalists, bronze medalists, and those who unfortunately do not win any medals. At the early stages of the competition, we cannot ascertain the true capabilities of participating athletes, thus we can only predict that the probabilities of these four outcomes for each athlete are equal.

Similarly, within the total medals awarded to each country, we use a multinomial distribution as the prior distribution, that is

$$(X, Y, Z|N) \sim Multinomial(N, p_1, p_2, p_3) \tag{3}$$

Where, p_1, p_2, and p_3 are the prior probabilities of gold, silver, and bronze medals, respectively; and $p_1 + p_2 + p_3 = 1$.

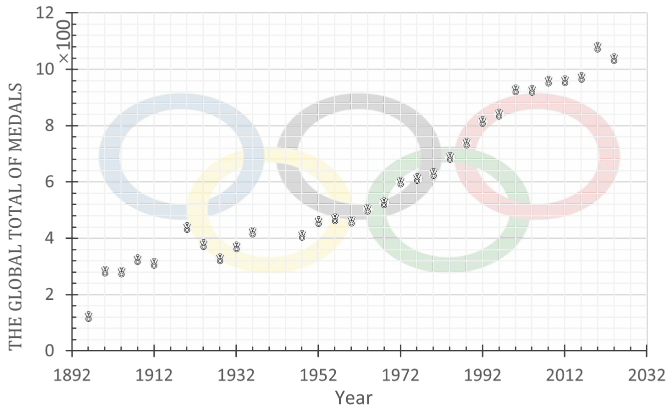


Figure 4: Global Total Olympic Medals Over Time

5.2.2 Model Establishment

Based on the specific medal counts of various countries, $D^i(t) = X^i(t), Y^i(t), Z^i(t)$ represents the number of gold, silver, and bronze medals country i wins in t year.

Here, $i \in \{France, UnitedStates, Germany, Australia, ...\}$, $t \in \{1896, 1900, ..., 2024\}$.

Assuming the observations for year t from country i are independent and identically distributed given the winning situation, the sample likelihood function can be expressed as $P(D^i|X^i, Y^i, Z^i) = \prod_t (p(x^i(t), y^i(t), z^i(t)|X, Y, Z)$.

Under the assumption of a multinomial prior distribution, for every group of historical sample data, $(x^i(t), y^i(t), z^i(t))$, its probability distribution can be expressed as $P(x^i(t), y^i(t), z^i(t)|N) = \frac{N!}{x^i!y^i!z^i!} P_1^{x^i} P_2^{y^i} P_3^{z^i}$, thus the likelihood function is $P(D|X, Y, Z) = \prod_{i=1}^{n} \frac{N!}{x^i!y^i!z^i!} P_1^{x^i} P_2^{y^i} P_3^{z^i}$.

Substituting the prior distribution and the likelihood function into the equation, and computing the normalization constant C ensures that $\sum_{x+y+z=N} P(X = x, Y = y, Z = z|D) = 1$.

Thus, the posterior distribution can be expressed as: $P(X^i(t), Y^i(t), Z^i(t)|D) = \frac{1}{c} \times \frac{N!}{x!y!z!} P_1^x P_2^y P_3^z \prod_{t=1}^{n} \frac{N!}{x^i(t)!y^i(t)!z^i(t)!} P_1^{x^i(t)} P_2^{y^i(t)} P_3^{z^i(t)}$.

By introducing the Bayesian model, in combination with the prior distribution and historical data, we can effectively evaluate the number of gold medals won given the total number of medals is known.

5.3 Model Improvement Based on Advantage Project Relaxation Function

As the medal count does not adequately measure the ability of athletes from countries that have never won any medals, we modify the multiple linear regression model. In this case, the assessment of athletes' capabilities from a country is not only based on the total count of medals won but also considers all data regarding the events and awards of athletes provided in the data set, including results from unofficial competitions. Thus, we adjust the independent variable $P\_Num^i(t)$ to ensure $P_1\_Num^i(t) = P + P\_Num^i(t)$, where P indicates the minimum level of the country's athletes, measured by the participation rate in events from that country. Additionally, the effect of project advantages also influences the medal-winning outcomes of countries to some extent. By considering the number and types of events, we adjust the independent variable $S\_Num^i(t)$ as follows:

$$S_1\_Num^i(t) = S\_Num^i(t) + f(p) \tag{4}$$

Where, $f(p) = \sum_{k=1}^{n} w_k I_k$ and w_k represent the weight proportions for various projects; I_k represents the count variable for project types.

Therefore, the modified multiple linear regression function is $MP^i(t,\theta) = a_1^i + a_2^i \times S_1\_Num^i(t) + a_3^i \times P_1\_Num^i(t) + a_4^i \times H^i(t) + a_5^i MP^i(t-4)$.

Since the previously mentioned random variables $X^i(t)$, $Y^i(t)$, and $Z^i(t)$ do not constitute a completely independent variation process in practice, their random fluctuations and magnitudes are influenced by the types and number of events that the country participates in, thus the values of $P_1^i(t)$, $P_2^i(t)$, and $P_3^i(t)$ will change with the number and type of events, as categorized below:

$$Item_{i,j}(t) = \frac{M_{i,j}(t)}{S_{i,j}(t)} \times \frac{P\_Num^i(t)}{S\_Num^i(t)} \tag{5}$$

Advantage Projects: ??? Balanced Projects:??? Disadvantage Projects:???

Figure 5: Advantageous projects and corresponding countries

In this thesis, we observe that when a project is an advantage project for the country, there is a higher likelihood of winning gold medals, while performance in non-advantage projects is less optimistic. Therefore, we introduce an advantage project relaxation function $f(M)$ to correct the Bayesian model $P_j^{oi}(t) = P_j^i(t)f(M)$.

Thus, the revised Bayesian estimation is: P_1^o

P_1

$Me^i \times P_1^{oi}$ This allows us to estimate the value of p1 through Bayesian estimation, ultimately predicting the number of gold medals won by various countries to be (variable XXX).

$$\begin{cases} m = min\left\{k \in N : 1 - \frac{P(k+1,\lambda^0(t))}{P(k+1)} \leq \frac{\alpha}{2}\right\} \\ n = min\left\{k \in N : \frac{P(k,\lambda^0(t))}{P(k)} \leq \frac{\alpha}{2}\right\} \end{cases} \tag{6}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(F(t_i,\theta) - F(t_i,\hat{\theta}))^2}{\sum_{i=1}^{n}(F(t_i,\theta) - \bar{F}(t_i,\hat{\theta}))^2} \tag{7}$$

5.4 Uncertainty analysis of the influence model

e^i

MP_j^i

$\hat{MP}_j^i$

Table 4: The prediction of the award situations in 2028

Top 10			Bottom 10		
Country	rank	Medal	Country	rank	Medal
UnitedStates	1	106	Hungary	11	23
China	2	87	Cuba	12	22
GreatBritain	3	82	Canada	13	20
Germany	4	59	Ukraine	14	19
Australia	5	47	Brazil	15	18
Italy	6	42	Spain	16	18
Japan	7	42	Sweden	17	16
SouthKorea	8	35	Kenya	18	15
Netherlands	9	27	Norway	19	15
France	10	23	Poland	20	15

Table 5: The probability of achieving a zero breakthrough

Country	Probability	Country	Probability
Nicaragua	0.129217	Tuvalu	0.131417
Libya	0.112495	South Sudan	0.118437
Comoros	0.144345	Lebanon	0.131196
Malawi	0.130465	Democratic Republic of the Congo	0.145907
Rhodesia	0.131206	East Timor	0.102365
Cook Islands	0.100384	Eswatini	0.097714
Peri	0.137529	St Kitts and Nevis	0.116556
South Vietnam	0.140657	Grenadines	0.122315
Cambodia	0.148522	Timor-Leste	0.13819
Kingfisher	0.117098	Brunei Darussalam	0.097726
Yangon	0.129572	DR Congo	0.120408

Table 6: prediction intervals for confidence

alpha	min_m	min_n
0.05	29285	1
0.10	29232	1
0.25	29147	1

$$e_j^i = MP_j^i - \hat{MP}_j^i \tag{8}$$

$E(e_j^i) = 0$

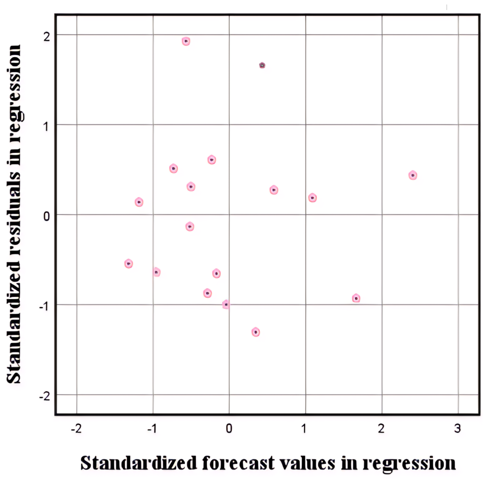


Figure 6: Residual plot

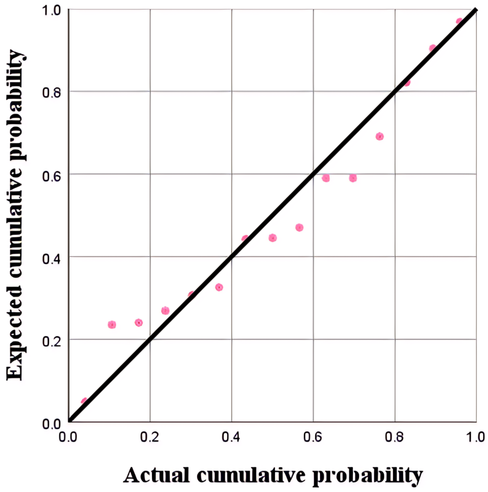


Figure 7: Quantile-Quantile plot

Since we try to keep the temperature of the hot water in bathtub to be even, we have to derive the amount of inflow water and the energy dissipated by the hot water into the air.

We derive the basic convection heat transfer control equations based on the former scientists' achievement. Then, we define the mean temperature of bath water. Afterwards, we determine two types of heat transfer: the boundary heat transfer and the evaporation heat transfer. Combining thermodynamic formulas, we derive calculating results. Via Fluent software, we get simulation results.

5.5 Revealing the results of the medal table for future Olympic Games

According to thermodynamics knowledge, we recall on basic convection heat transfer control equations in rectangular coordinate system. Those equations show the relationship of the temperature of the bathtub water in space.

We assume the hot water in the bathtub as a cube. Then we put it into a rectangular coordinate system. The length, width, and height of it is a, b and c.

In the basis of this, we introduce the following equations [**5**]:

- **Continuity equation:**

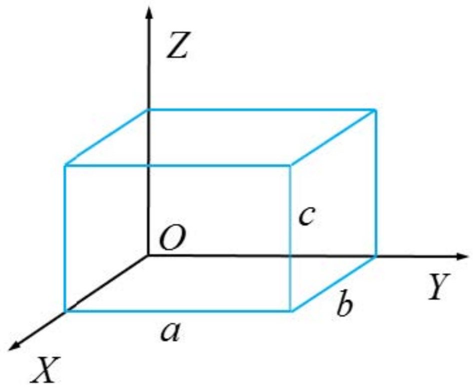


Figure 8: Modeling process

$$\frac{\partial u}{\partial x}+\frac{\partial v}{\partial y}+\frac{\partial w}{\partial z}=0 \tag{9}$$

where the first component is the change of fluid mass along the X-ray. The second component is the change of fluid mass along the Y-ray. And the third component is the change of fluid mass along the Z-ray. The sum of the change in mass along those three directions is zero.

- **Moment differential equation (N-S equations):**

$$\begin{cases} \rho\Big(u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}+w\frac{\partial u}{\partial z}\Big)=-\frac{\partial p}{\partial x}+\eta\Big(\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}+\frac{\partial^2 u}{\partial z^2}\Big) \\ \rho\Big(u\frac{\partial v}{\partial x}+v\frac{\partial v}{\partial y}+w\frac{\partial v}{\partial z}\Big)=-\frac{\partial p}{\partial y}+\eta\Big(\frac{\partial^2 v}{\partial x^2}+\frac{\partial^2 v}{\partial y^2}+\frac{\partial^2 v}{\partial z^2}\Big) \\ \rho\Big(u\frac{\partial w}{\partial x}+v\frac{\partial w}{\partial y}+w\frac{\partial w}{\partial z}\Big)=-g-\frac{\partial p}{\partial z}+\eta\Big(\frac{\partial^2 w}{\partial x^2}+\frac{\partial^2 w}{\partial y^2}+\frac{\partial^2 w}{\partial z^2}\Big) \end{cases} \tag{10}$$

- **Energy differential equation:**

$$\rho c_p\Big(u\frac{\partial t}{\partial x}+v\frac{\partial t}{\partial y}+w\frac{\partial t}{\partial z}\Big)=\lambda\Big(\frac{\partial^2 t}{\partial x^2}+\frac{\partial^2 t}{\partial y^2}+\frac{\partial^2 t}{\partial z^2}\Big) \tag{11}$$

where the left three components are convection terms while the right three components are conduction terms.

By Equation (**??**), we have ......

......

On the right surface in Fig. **??**, the water also transfers heat firstly with bathtub inner surfaces and then the heat comes into air. The boundary condition here is ......

5.5.1 Definition of the Mean Temperature

......

5.5.2 Determination of Heat Transfer Capacity

......

5.6 Sub-model III: Adding Water Discontinuously

p_1

$Me^i \times p_1^i$

In order to establish the unsteady sub-model, we recall on the working principle of air conditioners. The heating performance of air conditions consist of two processes: heating and standby. After the user set a temperature, the air conditioner will begin to heat until the expected temperature is reached. Then it will go standby. When the temperature get below the expected temperature, the air conditioner begin to work again. As it works in this circle, the temperature remains the expected one.

Inspired by this, we divide the bathtub working into two processes: adding hot water until the expected temperature is reached, then keeping this condition for a while unless the temperature is lower than a specific value. Iterating this circle ceaselessly will ensure the temperature kept relatively stable.

5.7 Heating Model

5.7.1 Control Equations and Boundary Conditions

5.7.2 Determination of Inflow Time and Amount

5.8 Standby Model

5.9 Results

We first give the value of parameters based on others' studies. Then we get the calculation results and simulating results via those data.

5.9.1 Determination of Parameters

After establishing the model, we have to determine the value of some important parameters.

As scholar Beum Kim points out, the optimal temperature for bath is between 41 and 45°C [1]. Meanwhile, according to Shimodozono's study, 41°C warm water bath is the perfect choice for individual health [2]. So it is reasonable for us to focus on 41°C $\sim$ 45°C. Because adding hot water continuously is a steady process, so the mean temperature of bath water is supposed to be constant. We value the temperature of inflow and outflow water with the maximum and minimum temperature respectively.

The values of all parameters needed are shown as follows: .....

5.9.2 Calculating Results

Putting the above value of parameters into the equations we derived before, we can get the some data as follows:

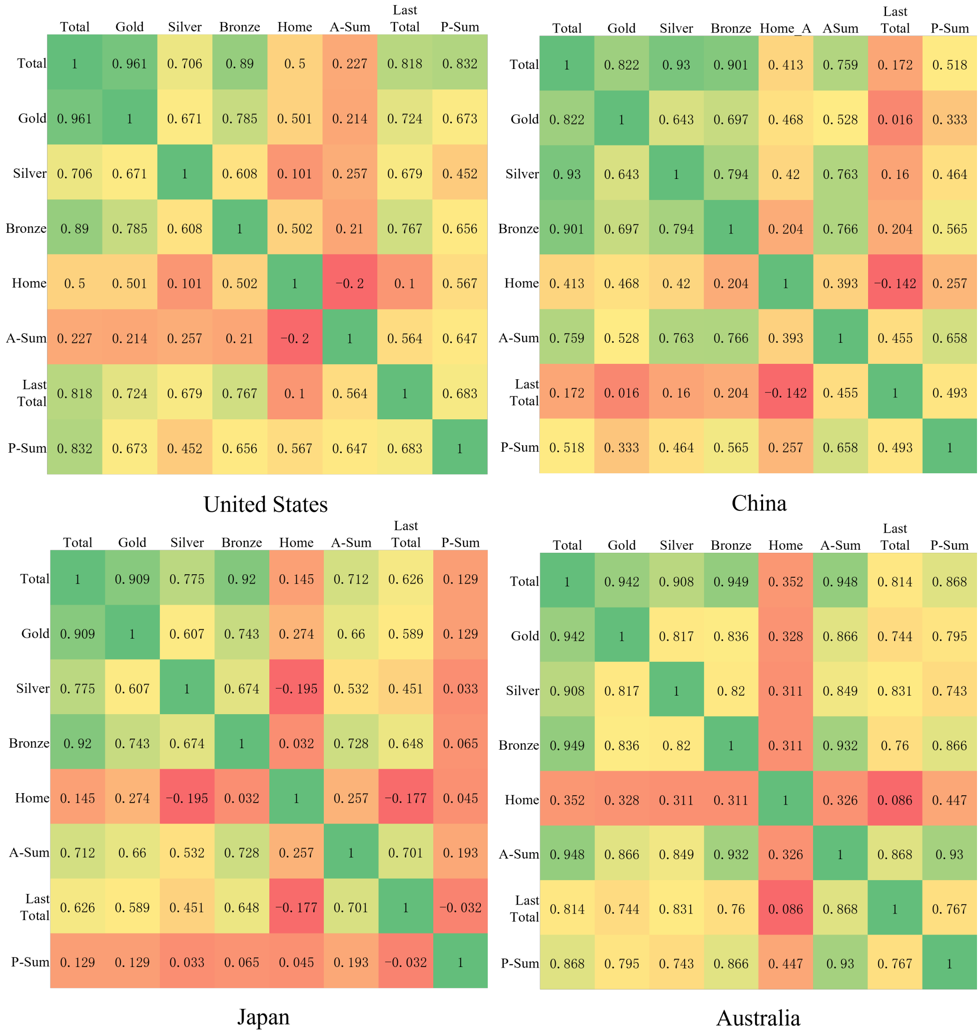


Figure 9: Correlation between the number of Total Medal and other factors

Table 7: The calculating results

Variables	Values	Unit
A_1	1.05	m^2
A_2	2.24	m^2
Φ_1	189.00	W
Φ_2	43.47	W
Φ	232.47	W
q_m	0.014	g/s

From Table **??**, ...

......

6 Correction and Contrast of Sub-Models

$$P_{i,j}(t) = M_{i,j}(t)/S_{i,j}(t) \tag{12}$$

$P_{i,j}(t)$

$M_{i,j}(t)$

$S_{i,j}(t)$

$\Delta P_{i,j} = P_{i,j}(t_1) - P_{i,j}(t_0)$

$$\Delta P_{i,j} = \bar{P}_{i,j}^{after}(t_0) - \bar{P}_{i,j}^{before}(t_0) \tag{13}$$

$$\Delta M_{i,j}(t) = \Delta P_{i,j} \times A_{i,j}(t) \tag{14}$$

$\Delta P_{i,j}$

$$MP^i = a_1^i + a_2^i \times S_1\_Num^i + a_3^i \times P_1\_Num^i + a_5^i M \tag{15}$$

$P_1\_Num^i = P_{ij} + \Delta P_{i,j} + P\_Num^i$

$$\Delta M = M_c - M_b \tag{16}$$

After establishing two basic sub-models, we have to correct them in consideration of evaporation heat transfer. Then we define two evaluation criteria to compare the two sub-models in order to determine the optimal bath strategy.

6.1 Correction with Evaporation Heat Transfer

Someone may confuse about the above results: why the mass flow in the first sub-model is so small? Why the standby time is so long? Actually, the above two sub-models are based on ideal conditions without consideration of the change of boundary conditions, the motions made by the person in bathtub and the evaporation of bath water, etc. The influence of personal motions will be discussed later. Here we introducing the evaporation of bath water to correct sub-models.

6.2 Contrast of Two Sub-Models

Firstly we define two evaluation criteria. Then we contrast the two submodels via these two criteria. Thus we can derive the best strategy for the person in the bathtub to adopt.

7 Model Sensitivity Analysis

7.1 The Influence of Different Bathtubs

Definitely, the difference in shape and volume of the tub affects the convection heat transfer. Examining the relationship between them can help people choose optimal bathtubs.

7.1.1 Different Volumes of Bathtubs

In reality, a cup of water will be cooled down rapidly. However, it takes quite long time for a bucket of water to become cool. That is because their volume is different and the specific heat of water is very large. So that the decrease of temperature is not obvious if the volume of water is huge. That also explains why it takes 45 min for 320 L water to be cooled by 1°C.

In order to examine the influence of volume, we analyze our sub-models by conducting sensitivity Analysis to them.

We assume the initial volume to be 280 L and change it by ±5%, ±8%, ±12% and ±15%. With the aid of sub-models we established before, the variation of some parameters turns out to be as follows

Table 8: Variation of some parameters

V	A_1	A_2	T_2	q_{m1}	q_{m2}	Φ_q
-15.00%	-5.06%	-9.31%	-12.67%	-2.67%	-14.14%	-5.80%
-12.00%	-4.04%	-7.43%	-10.09%	-2.13%	-11.31%	-4.63%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%
-8.00%	-2.68%	-4.94%	-6.68%	-1.41%	-7.54%	-3.07%

8 Task 2:

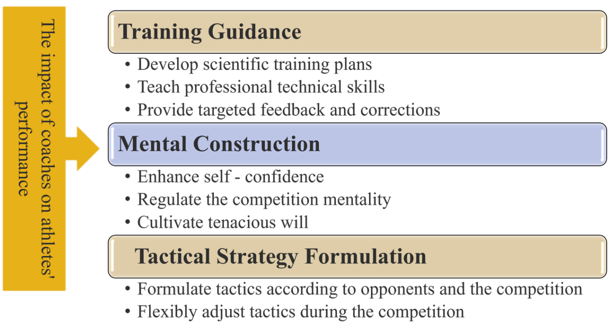


Figure 10: The impact of coaches on athletes' performance

$\Delta P_{i,j}$

Table 9: Table of variance analysis

Inspection value	Sample size	Standard deviation	z	P
1.12	25	2.538	2.43	0.015**
2.34	25	2.349	3.26	0.011**
2.12	24	1.956	2.81	0.019**

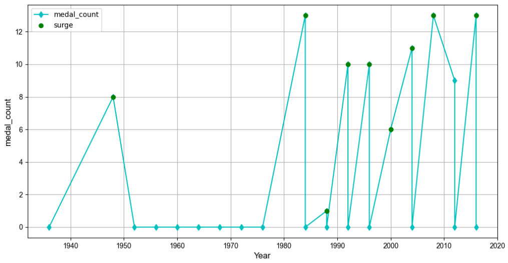


Figure 11: The number of Olympic medals won by the United States in women's gymnastics

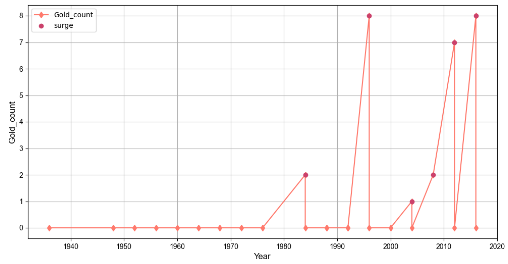


Figure 12: The number of Olympic gold medals won by the United States in women's gymnastics

8.1 Strengths

- We analyze the problem based on thermodynamic formulas and laws, so that the model we established is of great validity.
- Our model is fairly robust due to our careful corrections in consideration of real-life situations and detailed sensitivity analysis.
- Via Fluent software, we simulate the time field of different areas throughout the bathtub. The outcome is vivid for us to understand the changing process.
- We come up with various criteria to compare different situations, like water consumption and the time of adding hot water. Hence an overall comparison can be made according to these criteria.
- Besides common factors, we still consider other factors, such as evaporation and radiation heat transfer. The evaporation turns out to be the main reason of heat loss, which corresponds with other scientist's experimental outcome.

Table 10: Data files and encoding formats

Country	Pre_Mc_2028	Rank	Pre_Mb_2028	Rank	dete_M
Iran	5	46	8	32	3
Ireland	4	54	7	33	3
Uzbekistan	5	48	7	34	2

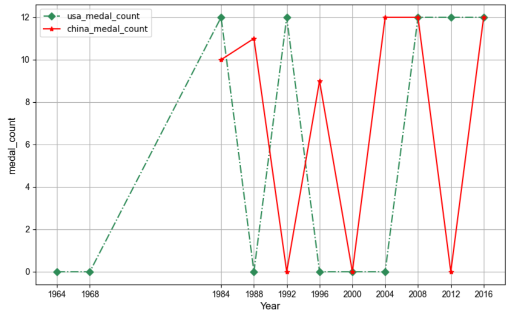


Figure 13: The number of Olympic medals won by China and the United States in women's volleyball

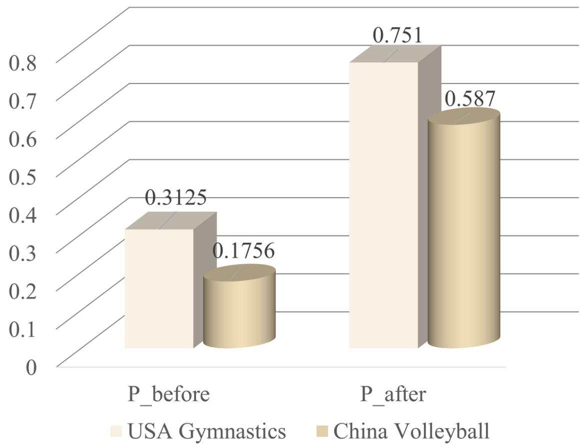


Figure 14: Description of the Differences

8.2 Weaknesses

- Having knowing the range of some parameters from others' essays, we choose a value from them to apply in our model. Those values may not be reasonable in reality.
- Although we investigate a lot in the influence of personal motions, they are so complicated that need to be studied further.
- Limited to time, we do not conduct sensitivity analysis for the influence of personal surface area.

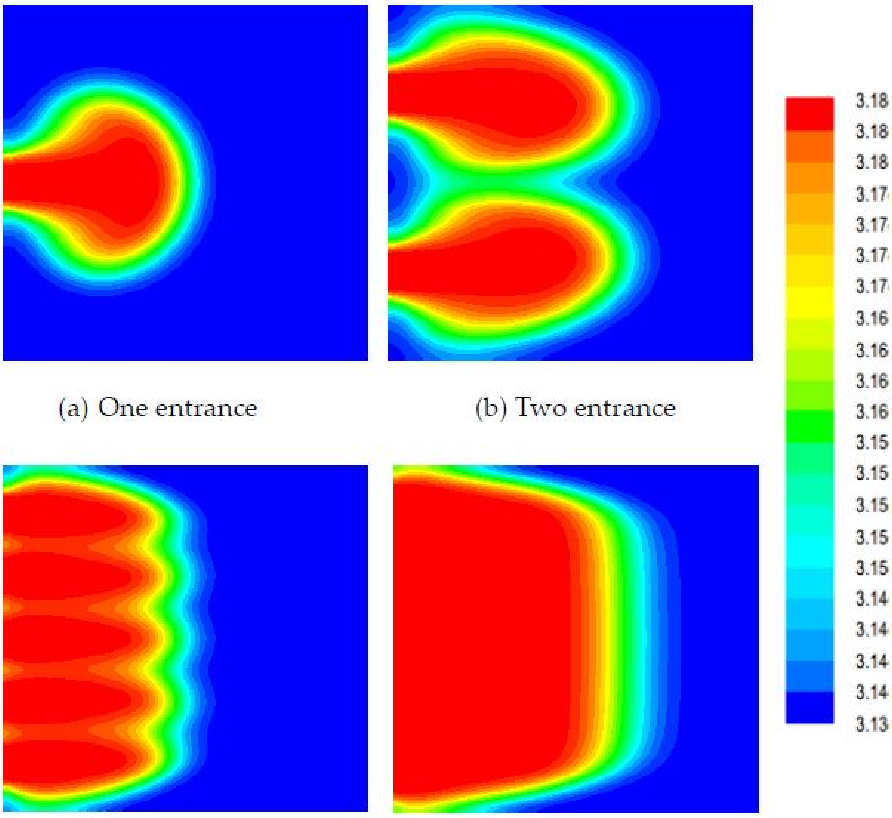
9 Further Discussion

In this part, we will focus on different distribution of inflow faucets. Then we discuss about the real-life application of our model.

- Different Distribution of Inflow Faucets

  In our before discussion, we assume there being just one entrance of inflow.

  From the simulating outcome, we find the temperature of bath water is hardly even. So we come up with the idea of adding more entrances.



(a) One entrance (b) Two entrance

(c) Five entrance (d) The whole side as entrance

Figure 15: The simulation results of different ways of arranging entrances

The simulation turns out to be as follows

From the above figure, the more the entrances are, the evener the temperature will be. Recalling on the before simulation outcome, when there is only one entrance for inflow, the temperature of corners is quietly lower than the middle area.

In conclusion, if we design more entrances, it will be easier to realize the goal to keep temperature even throughout the bathtub.

- Model Application

  Our before discussion is based on ideal assumptions. In reality, we have to make some corrections and improvement.

  1) Adding hot water continually with the mass flow of 0.16 kg/s. This way can ensure even mean temperature throughout the bathtub and waste less water.
  2) The manufacturers can design an intelligent control system to monitor the temperature so that users can get more enjoyable bath experience.
  3) We recommend users to add bubble additives to slow down the water being cooler and help cleanse. The additives with lower thermal conductivity are optimal.
  4) The study method of our establishing model can be applied in other area relative to convection heat transfer, such as air conditioners.

References

[1] Gi-Beum Kim. Change of the Warm Water Temperature for the Development of Smart Healthecare Bathing System. Hwahak konghak. 2006, 44(3): 270-276.

[2] https://en.wikipedia.org/wiki/Convective_heat_transfer#Newton.27s_law_of_cooling

[3] https://en.wikipedia.org/wiki/Navier%E2%80%93Stokes_equations

[4] https://en.wikipedia.org/wiki/Computational_fluid_dynamics

[5] Holman J P. Heat Transfer (9th ed.), New York: McGraw-Hill, 2002.

[6] Liu Weiguo, Chen Zhaoping, ZhangYin. Matlab program design and application. Beijing: Higher education press, 2002. (In Chinese)

Enjoy Your Bath Time!

From simulation results of real-life situations, we find it takes a period of time for the inflow hot water to spread throughout the bathtub. During this process, the bath water continues transferring heat into air, bathtub and the person in bathtub. The difference between heat transfer capacity makes the temperature of various areas to be different. So that it is difficult to get an evenly maintained temperature throughout the bath water.

In order to enjoy a comfortable bath with even temperature of bath water and without wasting too much water, we propose the following suggestions.

- Adding hot water consistently
- Using smaller bathtub if possible
- Decreasing motions during bath
- Using bubble bath additives
- Arranging more faucets of inflow

Sincerely yours,

Your friends

Appendices

Appendix A First appendix

In addition, your report must include a letter to the Chief Financial Officer (CFO) of the Goodgrant Foundation, Mr. Alpha Chiang, that describes the optimal investment strategy, your modeling approach and major results, and a brief discussion of your proposed concept of a return-on-investment (ROI). This letter should be no more than two pages in length.

Here are simulation programmes we used in our model as follow.

Input matlab source:

```
function [t,seat,aisle]=OI6Sim(n,target,seated)
pab=rand(1,n);
for i=1:n
    if pab(i)<0.4
        aisleTime(i)=0;
    else
        aisleTime(i)=trirnd(3.2,7.1,38.7);
    end
end
```

Appendix B Second appendix

some more text **Input C++ source:**

```
//===========================================================================
// Name        : Sudoku.cpp
// Author      : wzlf11
// Version     : a.0
// Copyright   : Your copyright notice
// Description : Sudoku in C++.
//===========================================================================

#include <iostream>
#include <cstdlib>
#include <ctime>

using namespace std;

int table[9][9];

int main() {

    for(int i = 0; i < 9; i++){
        table[0][i] = i + 1;
    }

    srand((unsigned int)time(NULL));

    shuffle((int *)&table[0], 9);

    while(!put_line(1))
```

```
    {
        shuffle((int *)&table[0], 9);
    }

    for(int x = 0; x < 9; x++){
        for(int y = 0; y < 9; y++){
            cout << table[x][y] << " ";
        }

        cout << endl;
    }

    return 0;
}
```